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Schiff-based polymer covalent adaptable networks with excellent tunable mechanical properties

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ABSTRACT

In dynamic covalent networks, the activation energy increases with the crosslink density of the network when the dynamic bond content is constant. In this study, only a small amount of N-[3-(trimethoxysilyl)propyl]ethylenediamine (ANN) was added to regulate the crosslink density of the dynamic crosslinked network without changing its activation energy. Biobased Schiff-based polymer networks, EAPA-X, were prepared using biodialdehyde (EAAP), Priamine 1071, and N-[3-(trimethoxysilyl)propyl]ethylenediamine (ANN). Comprehensive characterization, including nuclear magnetic resonance spectroscopy (NMR), thermogravimetric analysis (TGA), dynamic thermomechanical analysis (DMA), mechanical testing, and reprocessing measurements, was conducted on the prepared films. The results indicate that by varying the ANN content from 0 % to 10 %, EAPA-X films achieved tunable Young's modulus values ranging from 4 to 1034 MPa. This improvement can be attributed to the condensation behavior of methoxysilane groups in ANN, which enhances the crosslinking density of the polymer system due to its role as a hard chain segment. Interestingly, despite the significant difference in crosslinking density (340–1994 mol/m^3) between EAPA-X films, the activation energies remained similar, showing low values around 50 kJ/mol. Additionally, the films demonstrated excellent self-healing and recyclability. These results highlight the significant potential of EAPA-X films for sustainable applications in smart materials, electronic packaging, and flexible semiconductor materials.

1. Introduction

Thermoset polymers, also known as thermoset resins, form their crosslinked networks through chemical covalent bonding reactions, establishing crosslink points within the molecular structure. Due to their unique crosslinked architecture, thermoset polymers typically exhibit superior mechanical properties, dimensional stability, and thermal resistance compared to thermoplastic polymers. Consequently, they are widely used in aerospace, construction, composite materials, and other fields[1–3]. In 2011, Leibler et al. pioneered the introduction of dynamic covalent bonding into thermoset polymers, developing a novel material called vitrimer, which is classified as a type of covalent adaptable network (CAN)[4]. Similar to traditional thermosets, CANs possess crosslinked networks; however, these networks involve dynamic covalent bonds. These dynamic bonds can be activated to exchange under certain external conditions (e.g., light, heat, pressure), allowing the network topology to change while maintaining a constant crosslink density[5–7].

In recent years, significant progress has been made in the study of CANs, including various chemistries such as Diels–Alder cycloaddition [8], oxime ester formation[9], disulfide bonds[10], dynamic enamine bond[11,12], dynamic imine bonds[13], boronic ester exchange[14], Schiff bases[15], and others. Among these, Schiff bases[16] are particularly noteworthy because they can be easily synthesized from aldehydes or ketones and amines under mild conditions with high yields. In the context of Schiff-based CANs, biobased materials have attracted significant attention due to the abundance of raw materials and the excellent reprocessing properties of these systems. However, the current

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methods for preparing Schiff-based CANs from biobased amines often result in materials with poor mechanical properties or high self-healing temperatures[17–20]. For example, Dhers et al.[21] developed biobased Schiff base vitrimer materials by curing furfural with Priamine 1071, a mixture of biobased polyamines extracted from vegetable oils. While the resulting films demonstrated fast stress relaxation at room temperature, they exhibited poor mechanical strength (0.7 MPa). Similarly, Zhang et al.[22] synthesized biobased vitrimer materials with high strength by reacting biobased diamine (Priamine 1074) with hyperbranched biobased cyclocarbonate. Although the resulting films exhibited excellent mechanical properties (up to 35 MPa), the high self-healing temperature of 160 °C limited their practical application. These tradeoffs present significant challenges for the development of smart materials. The flexible structure of biobased materials can significantly reduce the crosslinking density of dynamic CANs, leading to performance limitations[23,24]. To address these issues, recent studies have proposed strategies such as physical compounding or the introduction of rigid structures to modulate material properties[25–27]. For example, Jiyae Hong et al.[28] enhanced the properties of Schiff base films by incorporating carbon fiber cloth through physical compounding. However, challenges related to the complex preparation process and the inability to precisely control the material properties remain unresolved. Bonab et al.[29] dispersed carbon nanotubes (CNTs) into polyurethane networks, achieving precise control over the film's mechanical properties, increasing the Young's modulus from 1.6 ± 0.2 MPa to 5.6 ± 0.4 MPa. However, this system still required heating to 160 °C for scratch repair, and the inherent defects of heterogeneous mixtures persisted. Compared to physical doping methods, chemical crosslinking offers greater stability, as it avoids issues related to film strength inconsistency caused by uneven preparation or fatigue damage due to interlayer separation. This makes chemical crosslinking a more promising approach for overcoming the limitations of current materials in the field of dynamic covalent networks.

In this study, a biodialdehyde monomer was solidified with the biobased polyamine Priamine 1071 and N-[3-(trimethoxysilyl)propyl] ethylenediamine (ANN). By controlling the ANN content, a series of EAPA-X CAN films were synthesized. Owing to the three-dimensional siloxane-based structure of ANN, the siloxane groups condense with each other, significantly enhancing the mechanical properties of the films. The results demonstrated that the tensile strength of the films ranged from 5.1 MPa to 22.3 MPa, and the elongation at break varied between 4 % and 202 % as the ANN content increased from 0 % to 10 %. This indicates that a rapid modulation of the films' mechanical properties can be achieved through slight adjustments in the ANN content. Among these films, EAPA-3 (containing 5 % ANN) exhibited the best overall mechanical performance, with a tensile strength of 16 MPa, an elongation at break of 53 %, and a toughness of 5.7 MJ/m^3. The activation energy of EAPA-3, determined via stress relaxation testing, was 50.7 kJ/mol, significantly lower than that of previously reported films with similar structures. Additionally, all EAPA-X films displayed fast exchange reactions of the dynamic imine bonds, with close activation energies at elevated temperatures, despite the significant differences in their crosslinking densities. These findings confirm that EAPA-X films possess tunable mechanical properties, excellent self-healing capabilities, recyclability, and hold great potential for applications in smart materials.

2. Experimental section

2.1. Materials

1,6-Hexanediol, propargylic acid, N-[3-(trimethoxysilyl)propyl]ethylenediamine, p-toluenesulfonic acid hydrate ($TsOH{\cdot}H_2O$), deuterated chloroform ($CDCl_3$), p-hydroxybenzaldehyde (PHB), N,N-dimethylformamide (DMF), and 1,4-diazabicyclo[2.2.2]octane (DABCO) were purchased from Energy Chemical. Priamine 1071 (industrial grade) was obtained from CRODA Company. Sodium chloride (NaCl) was sourced from Tianjin Hengxing Chemical Reagent Manufacturing Co., Ltd., and sodium bicarbonate ($NaHCO_3$) was purchased from Tianjin Zhiyuan Chemical Reagent Co., Ltd. Magnesium sulfate ($MgSO_4$) was acquired from Alab (Shanghai) Chemical Technology Co., Ltd., and chloroform was purchased from Xilong Technology Co., Ltd. Deionized water was produced in the laboratory. All materials were used as received without further purification, and the purity of all chemicals was above 95 %.

2.2. Synthesis of 1, 6-hexadiol dipropiolate

1,6-Hexanediol (5.68 g, 48 mmol), $TsOH{\cdot}H_2O$ (18.24 g, 96 mmol), propargylic acid (6.72 g, 96 mmol), and 50 mL chloroform were added to a 250 mL single-neck round-bottom flask equipped with a reflux condenser. The reaction mixture was refluxed at 50 °C for 18 h. The resulting solution was subsequently washed with deionized water, saturated $NaHCO_3$ solution, and saturated NaCl solution. The organic phase was dried over anhydrous $MgSO_4$, followed by recrystallization to obtain the final product.(See Schemes 1)

2.3. Synthesis of 4,4- (1,6-hexanediol dienone) dibenzaldehyde

TMMP (0.5 g, 2.25 mmol), PHB (0.55 g, 4.5 mmol), and DABCO (0.5 mol% relative to the hydroxyl groups of PHB, 0.026 mg, 0.023 mmol) were dissolved in 10 mL DMF at room temperature and stirred for 8 h. Ethanol was then added in a volume ratio of 8:1 (V_{EtOH}: V_{DMF} = 8:1), and the mixture was stirred at 80 °C for 30 min. The product was filtered, and the remaining DMF was removed by vacuum evaporation in an oil bath at 140 °C, yielding a yellow powder with an 83 % yield.(See Schemes 2)

2.4. Preparation of polymer network

The synthesis of polymer networks EAPA-X is outlined in Scheme 3 **and** Table 1. The general procedure is as follows: EAAP (1.0 equivalent), amine curing agents (1.05 equivalents), and 10 mL DMF were combined in a 20 mL round-bottom flask at room temperature and stirred for 2 h to form a homogeneous solution. The mixture was then poured into a polytetrafluoroethylene mold (8 cm × 8 cm × 1.5 cm) and cured in an oven at 120 °C for 24 h, followed by 150 °C for 5 h to obtain the final polymer films. The amine curing agents used were ANN and Priamine 1071. The films were prepared with varying ANN content, corresponding to 0 %, 2.5 %, 5 %, 7.5 %, and 10 % of the total amine functional groups in the curing agents.

2.5. Characterization

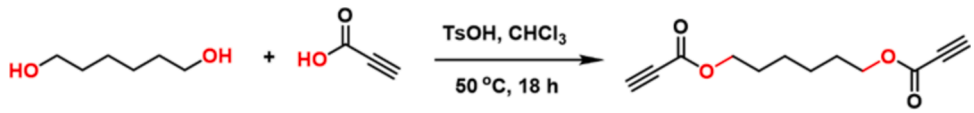
2.5.1. Structural analysis

The ^1H and ^{13}C nuclear magnetic resonance (NMR) spectra of the products were obtained using a 400 MHz or 100 MHz spectrometer (Bruker, Germany) with DMSO-d_6 and $CDCl_3$ as solvents.

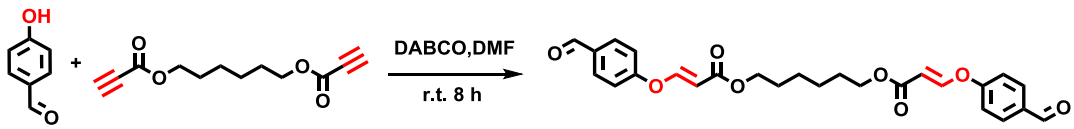
2.5.2. Thermodynamic, mechanical, and stress relaxation properties

Thermogravimetric analysis (TGA) was performed using the TA-Q50 instrument (New Castle, USA) over a temperature range of 25 °C to 750 °C under a nitrogen flow of 50 mL/min, with a heating rate of 10 °C/min. Dynamic mechanical analysis (DMA) was conducted from − 30 °C to 100 °C at a heating rate of 3 °C/min in stretch mode at a frequency of 1 Hz using the TA-Q800 instrument (New Castle, USA). The storage modulus (E'), loss modulus (E"), and tan δ (E'/E") were recorded at the respective temperatures. The crosslinking density (ν_e) of the glassy compound sample was calculated using the equation[30–33]:

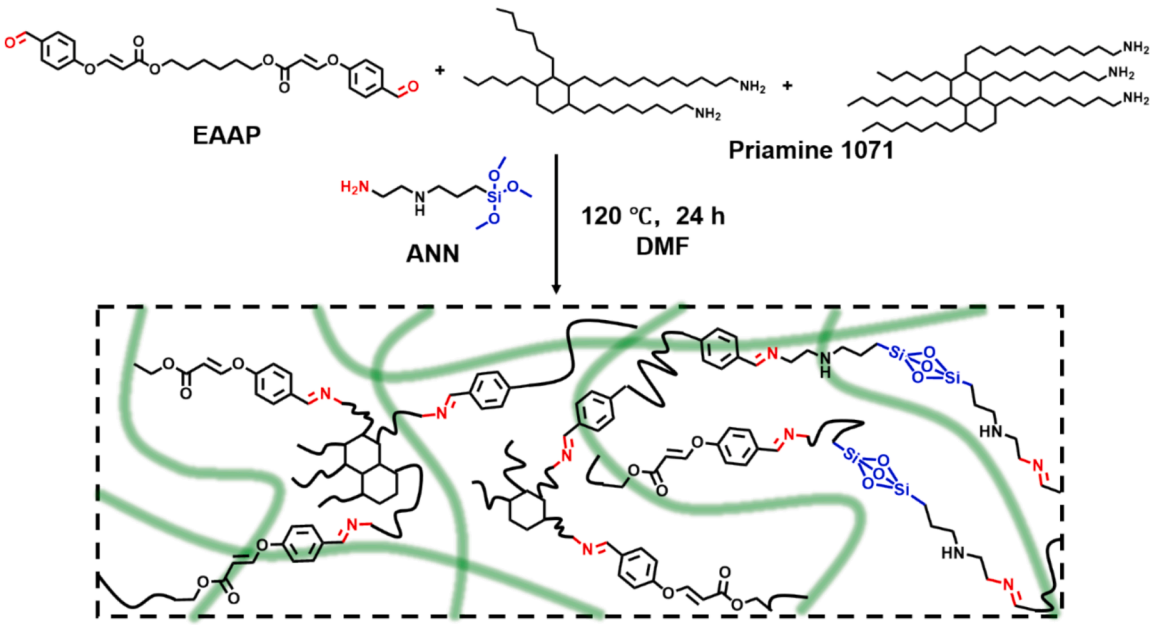
$$\nu_e = E'/3RT$$



Scheme 1. Synthesis of 1,6-hexanediol dipropiolate (TMMP).



Scheme 2. Synthesis of 4,4-(1,6-hexanediol dienone) dibenzaldehyde (EAAP).



Scheme 3. Preparation of polymer networks EAPA-X.

Table 1
Raw material ratio for preparing the CANs.

Sample	EAAP	ANN	Priamine 1071
EAPA-1	4.5 mmol	0 mmol	0.95 mmol
EAPA-2	4.5 mmol	0.12 mmol	0.92 mmol
EAPA-3	4.5 mmol	0.24 mmol	0.90 mmol
EAPA-4	4.5 mmol	0.35 mmol	0.87 mmol
EAPA-5	4.5 mmol	0.47 mmol	0.85 mmol

where R is the ideal gas constant, T is the temperature in Kelvin, and E' is the energy storage modulus at the test temperature (50 °C above T_g).

Stress relaxation analysis was also performed on the TA-Q800. The samples were stabilized at a set test temperature (ranging from 30 °C to 600 °C), followed by the application of a constant strain of 0.3 % for 20 min to monitor stress changes. The relaxation time (τ^*) was calculated using the Arrhenius equation[34,35]:

$$\tau^* = \tau_0{}^* exp_{RT}^{E_a}$$

where τ^* is the time recorded when the sample relaxes to 1/e of its initial modulus, τ_0 is the time recorded at infinite temperature, T is the temperature in Kelvin, R is the ideal gas constant, and E_a is the activation energy.

2.5.3. Frequency sweep test

Frequency sweep tests were conducted using the Discovery HR-2 Rheometer (New Castle, USA) over a frequency range of 0.1–100 rad/s, and at temperatures ranging from 110 °C to 140 °C.

2.5.4. Swelling ratio and gel content

For the swelling test, the membrane samples were first weighed (denoted as M_0) and then soaked in acetone at room temperature for 24 h. After reaching swelling equilibrium, the solvent was wiped off the surface, and the swollen weight of the sample was recorded as M_1. Finally, the sample was dried in an oven at 80 °C to a constant weight (M_2). The swelling ratio (SR) was calculated as[36]

$$SR = 1 - \frac{M_0 - M_1}{M_0}$$

The gel content (GL) was calculated as

$$GL = M_2/M_0$$

2.5.5. Self-healing and recovery experiments

The self-healing properties of the samples were evaluated using the CIMY-310 metallography microscope (COSSIM, Beijing). In this experiment, membrane samples were deliberately scraped, and the healing process was observed under the microscope. Optical images of the samples were captured at 1, 2, and 4 min, respectively. The self-healing

properties and efficiency were further assessed through stress–strain measurements of the healed samples. The self-healing efficiency (η) was calculated using the following equation[37]:

$$\eta = \frac{\sigma_{healed}}{\sigma_{origin}}$$

where σ_{healed} is the average tensile strength of the healed sample, and σ_{origin} is the average tensile strength of the original sample.

The recovery test was performed using a RYJ-600C hot press (Shanghai, China). All samples were first cut into small pieces and then hot pressed at 70 °C under 5 MPa for 15 min to obtain the recycled films.

3. Results and discussion

3.1. Characterization of the biobased dialdehyde monomers

The ^{1}H NMR and ^{13}C NMR spectra of TMMP, shown in **Figs. S1 and S2**, confirm the successful synthesis of TMMP. EAAP was prepared through an alkyne–phenol "click" reaction between PHB and TMMP, with the ^{1}H NMR spectrum of EAAP presented in Fig. 1**(a)**. The peak at 10.04 ppm corresponds to the proton of the aldehyde group (–CHO). The peaks at 8.01–7.96 ppm and 7.91–7.86 ppm are attributed to the protons of the aromatic ring. The peaks at 7.32–7.29 ppm and 5.81–5.76 ppm are assigned to the protons of the carbon–carbon double bonds (–CH = CH–). Additionally, the peaks at 4.26–4.19 ppm, 1.78–1.73 ppm, and 1.51–1.46 ppm are associated with the protons of alkyl groups (–CH_2-). The ^{13}C NMR spectrum of EAAP is shown in Fig. 1**(b)**, and it further supports the successful preparation of the biobased dialdehyde monomers. The characteristic peak for the −O-CH = group is located at 156.47 ppm, the characteristic peaks for the –CH = CH– groups are located at 134.15 ppm and 118.27 ppm [32], the characteristic peaks for the (–CH_2-) groups are located at 65.21 ppm, 29.88 ppm and 26.13 ppm, and the characteristic peaks for the −O-C = O- group is located at 105.39 ppm. [32] These spectral changes provide clear evidence of the successful synthesis of the biobased dialdehyde monomers EAAP.

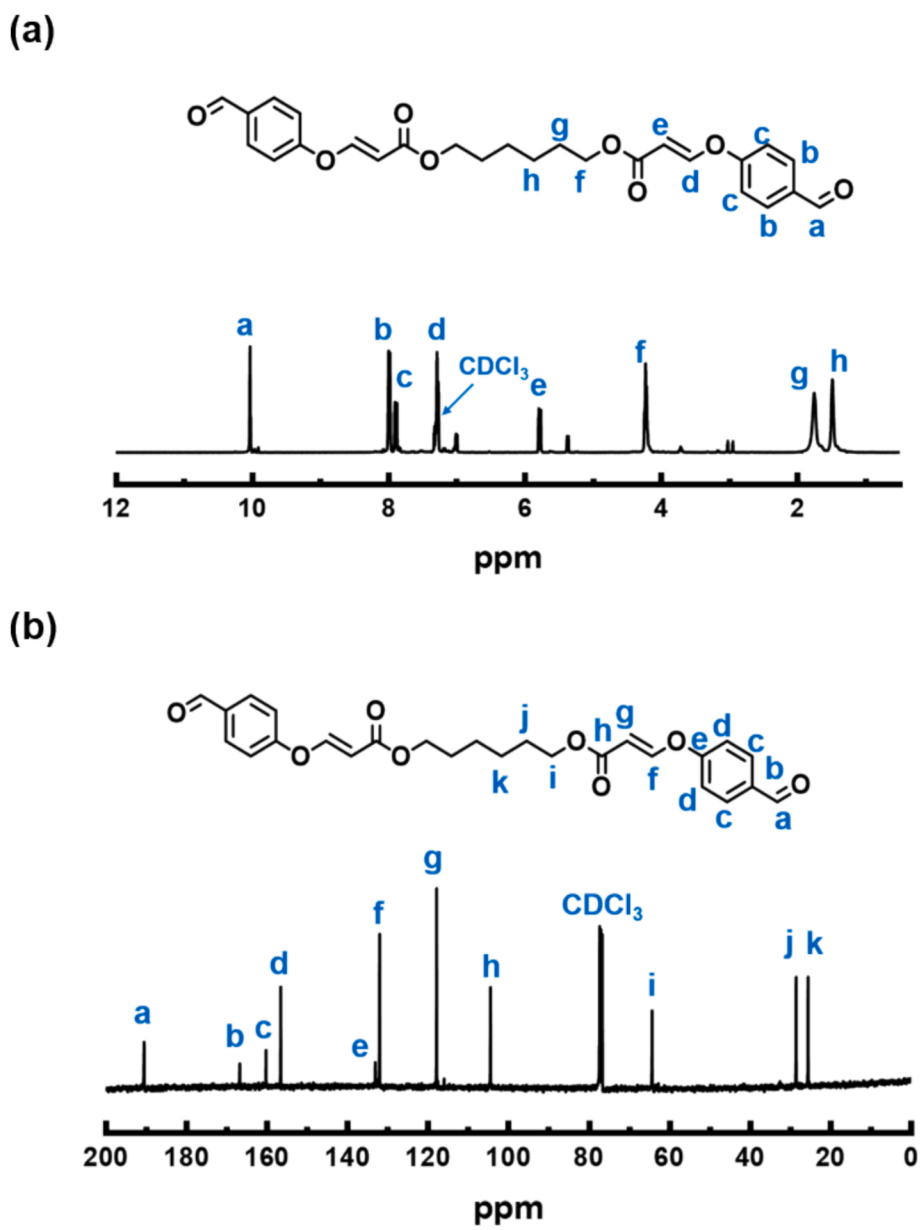


Fig. 1. ^{1}H NMR (a) and ^{13}C NMR (b) spectra of EAAP.

3.2. Mechanical and thermodynamic properties of EAPA-X

Mechanical properties are crucial in determining the performance of polymer networks. As shown in Fig. 2**(a–b)** and summarized in Table 2, EAPA-X films exhibit excellent tensile strength, ranging from 5.1 MPa to 22.3 MPa, with elongation at break varying from 202 % to 4 %. In addition, the calculated Young's modulus and toughness span from 4 MPa to 1034 MPa and 0.17 MJ/m^3–5.7 MJ/m^3, respectively. The significant variation in these mechanical properties underscores the superior tunability of the films, allowing for customization of functionality based on environmental requirements. Thermal properties of the polymer films, presented in **Fig. S3**, revealed two distinct weight-loss stages for all samples. The first stage, observed between 315 °C and 410 °C, corresponds to the dissociation of the Schiff base structure[38], while the second stage, occurring above 410 °C, is attributed to the degradation of the polymer backbone[39]. Dynamic mechanical properties were assessed through DMA tests, as shown in Fig. 2**(c–d)**. The glass transition temperatures (T_g) for the EAPA-X films were 43 °C, 53 °C, 65 °C, 80 °C, and 94 °C, respectively. Notably, all samples exhibited a distinct rubbery plateau at temperatures above T_g, which enabled the calculation of the crosslink density of the films. The crosslink density of the EAPA-X membranes increased significantly, ranging from 340 mol/m^3 to 1994 mol/m^3 as the siloxane content increased. The swelling measurements (**Fig. S4 and Table S1**) indicated an increase in gel content from 97 % to 99 %, while the SR decreased from 43 % to 26 % with the increasing degree of crosslinking in the EAPA-X films. The rise in T_g and crosslink density is primarily due to the increasing siloxane group content in the system. Notably, the T_g more than doubled when the siloxane monomer content reached 10 %. The condensation of methoxy and silicon groups is likely the primary contributor to the enhanced rigidity and crosslink density of the networks. As the proportion of hard segments increases, the crosslinked networks become stiffer and more brittle, while an increased proportion of soft segments enhances flexibility and reduces rigidity. This finding demonstrates that the tunability of mechanical properties in CANs can be effectively achieved by varying the ratio of soft to hard segments.

Table 2
Mechanical and thermal properties of the EAPA-X.

Simple	Stress at breaks (MPa)	Elongation at breaks (%)	T_g (DMA) (°C)	ν_e (mol/m^3)
EAPA-1	5.1	203	43	340
EAPA-2	8.6	102	53	618
EAPA-3	16.1	54	65	1140
EAPA-4	22.3	11	80	1861
EAPA-5	18.4	4	94	1994

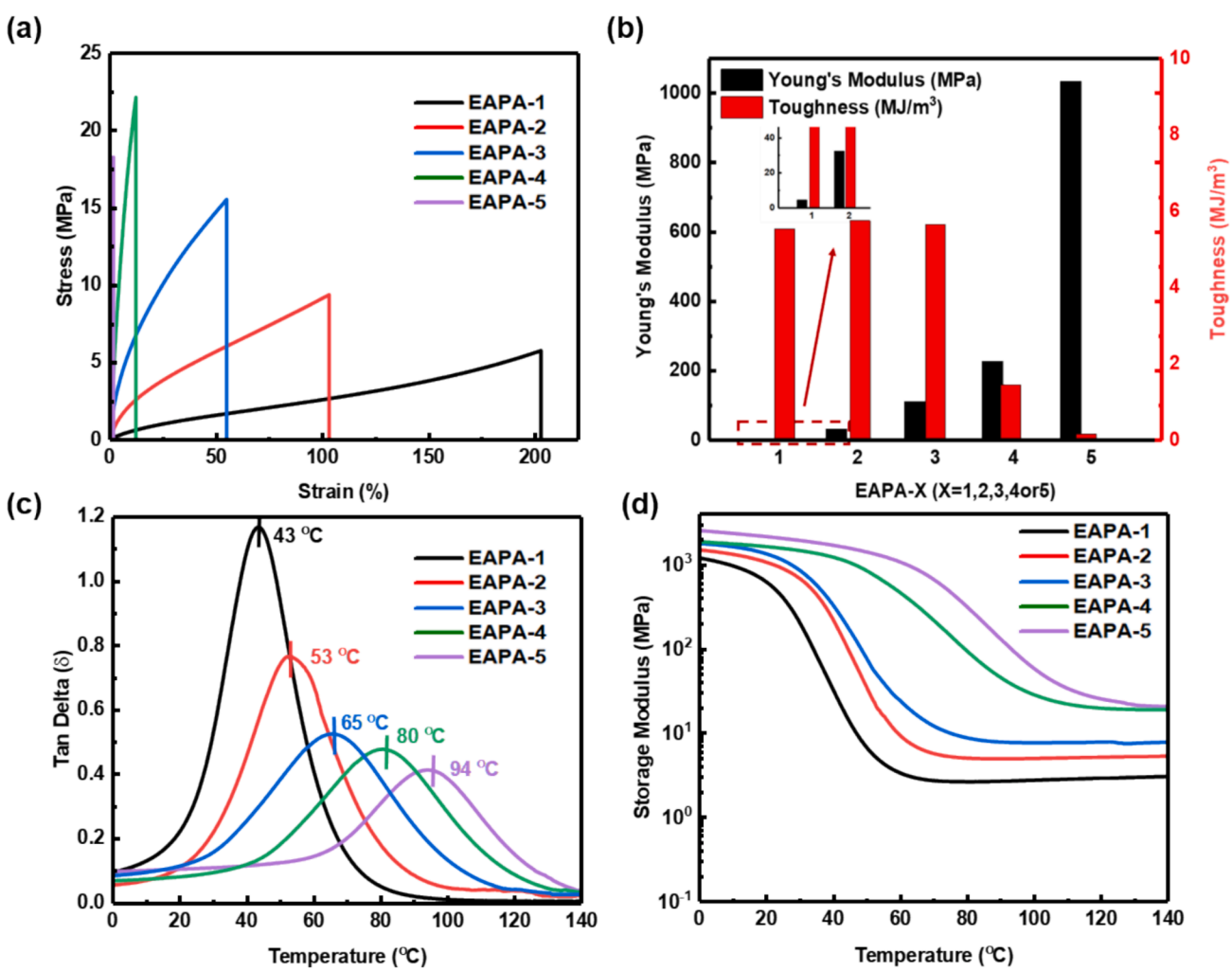


Fig. 2. (a) Stress–strain curve of EAPA-X, (b) Young's modulus and toughness of EAPA-X, (c–d) DMA curve of EAPA-X films.

3.3. Stress relaxation performance EAPA-X films

The dynamic exchange properties and viscoelastic behavior of the polymer films under varying temperature and deformation conditions were evaluated using stress relaxation measurements[40]. Since EAPA-3 exhibited the highest toughness and favorable mechanical strength, it was chosen as the representative sample for stress relaxation investigation, as shown in Fig. 3**(a)**. When the temperature increased from 40 °C to 70 °C, the relaxation time of EAPA-3 decreased significantly from 420 s to 97 s. This rapid reduction in relaxation time is attributed to the accelerated exchange reaction of dynamic imine bonds at elevated temperatures, enabling the polymer network to swiftly rearrange its topology, thereby exhibiting excellent stress relaxation properties. The stress relaxation behavior of other EAPA-X films is presented in **Fig. S5**, where their relaxation times were notably longer compared to EAPA-3.

To further investigate the stress relaxation properties of the EAPA-X films, the Arrhenius equation[41] was applied, as shown in Fig. 3**(b)**, to calculate the activation energy (E_a). The E_a values for EAPA-1, EAPA-2, EAPA-3, EAPA-4, and EAPA-5 were 49.7 kJ·mol^{-1}, 50.3 kJ·mol^{-1}, 50.7 kJ·mol^{-1}, 50.2 kJ·mol^{-1}, and 50.7 kJ·mol^{-1}, respectively. The similar Ea values across the EAPA-X films indicate that, unlike general crosslinked networks with constant dynamic bond content, these films do not exhibit a significant increase in activation energy as crosslink density increases. This observation confirms the presence of fast dynamic bond exchanges within the polymer networks, specifically the rapid exchange of imine bonds in the crosslinked structure. In addition, oscillation frequency sweep tests were performed over a temperature range of 110 °C to 140 °C, as shown in **Fig. S6**. A clear trend of decreasing energy storage modulus with increasing temperature was observed for all EAPA-X films. This decrease is attributed to variations in crosslink density caused by the dissociation of dynamic covalent bonds at elevated temperatures, further demonstrating that the dynamic covalent bonding in these films follows a dissociation-type mechanism.

3.4. Self-healing property of the polymer EAPA-X

The self-healing capability is a critical feature of polymer films, significantly impacting their practical applications[42]. Among the EAPA-X films, EAPA-3 was selected for further study due to its superior mechanical properties and toughness. Thanks to the reversible exchange reaction of dynamic covalent bonds, the crosslinked EAPA-X films exhibit self-healing behavior. As illustrated in Fig. 4**(a)**, when dynamic covalent bonds in the polymer film break (e.g., due to scratches), these bonds can rearrange and reform at elevated temperatures, allowing the film to self-heal at a macroscopic level. Tensile-strain tests of the self-healed EAPA-3 film at different times and temperatures are shown in **Fig. S7.** The self-healing efficiency of EAPA-3 is highly temperature-dependent, with the most significant improvement observed at 70 °C. Fig. 4**(b)** shows the stress–strain curves of EAPA-X films after 5, 10, 20, and 30 min of self-healing at 70 °C. After just 5 min at 70 °C, the EAPA-3 film achieved a self-healing efficiency of over 50 %, and after 20 min, the efficiency reached 85.4 %. Fig. 4**(c)** presents the self-healing efficiency of the EAPA-X films as a function of time at 70 °C. Notably, the EAPA-3 film, with its superior mechanical properties, attained an impressive self-healing efficiency of more than 85 % after 20 min at 70 °C. Other EAPA-X films also achieved close to 85 % efficiency under the same conditions. As shown in Table 3, these findings suggest that EAPA- X films demonstrate excellent self-healing properties compared to other similar systems. The consistent self-healing performance across the EAPA-X films (as shown in **Fig. S8**) can be attributed to the constant dynamic bond content in the crosslinked network. It is noteworthy that despite their relatively high T_g, the EAPA-X films still display significant self-repair efficiency, which is likely due to the low energy barriers associated with dynamic covalent bond exchange, as previously demonstrated.

Fig. 4**(d)** provides a visual representation of the self-healing process using optical electron microscopy. Initially, scratches with a width of 84 μm were clearly visible on the EAPA-3 film surface. Remarkably, after 20 min of self-healing, the scratches had nearly vanished, with only negligible traces remaining. Fig. 4**(e)** shows a dumbbell-shaped EAPA-3 film that was cut in the middle and subsequently rejoined. After 40 min of healing at 90 °C without the use of solvents or pressure, the reformed EAPA-3 film was fully intact and smooth. The welded film was even capable of suspending a weight of up to 200 g without breaking, demonstrating the exceptional self-healing properties of the EAPA-X films. Optical characterization of other EAPA-X films, as shown in **Fig. S9**, further confirms their excellent self-healing capabilities.

3.5. Reprocessing properties of the EAPA-X films

The reprocessing properties of the EAPA-X films were evaluated by conducting stress–strain tests on the samples before and after reshaping. As shown in Fig. 5**(a)**, the mechanical stress loss in the EAPA-3 film after the third cycle of reshaping is minimal, with retention of over 86 % compared to the original film. The remolding process of the EAPA-3 film is illustrated in Fig. 5**(b)**, where the film was cut and pressed at 70 °C under 5 MPa for 15 min. After three recycling cycles, the film remained flat and uniform, demonstrating its excellent reprocessing capabilities. **Fig. S10** shows the stress–strain curves of the EAPA-X films in their original form and after three cycles of recovery. The results indicate that all the recycled EAPA-X film samples maintained high recycling efficiency, confirming the promising reprocessability of the EAPA-X films. In order to evaluate the retention of the thermodynamic properties of the films, we performed DMA and DSC tests on EAPA-3 and its samples

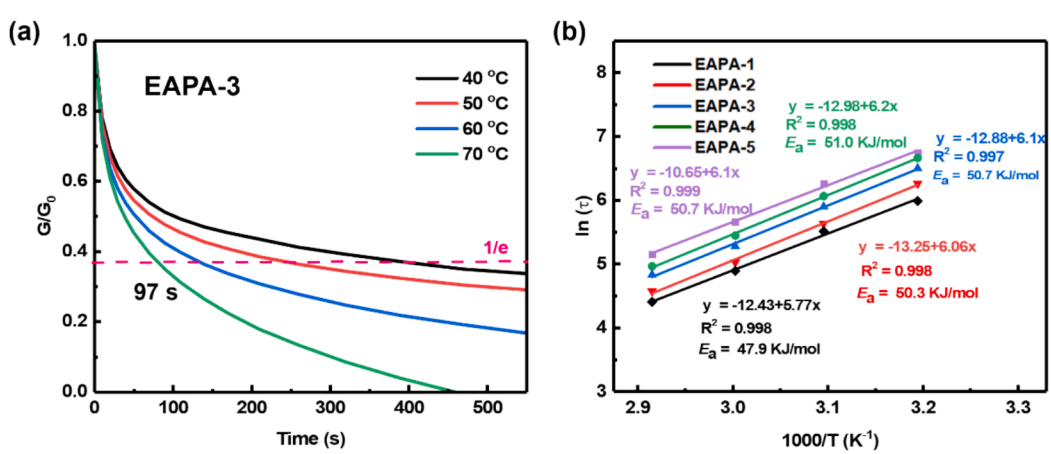


Fig. 3. (a) Stress relaxation curve of EAPA-3 film. (b) Activation energy of EAPA-X films, with Arrhenius fitting inserted.

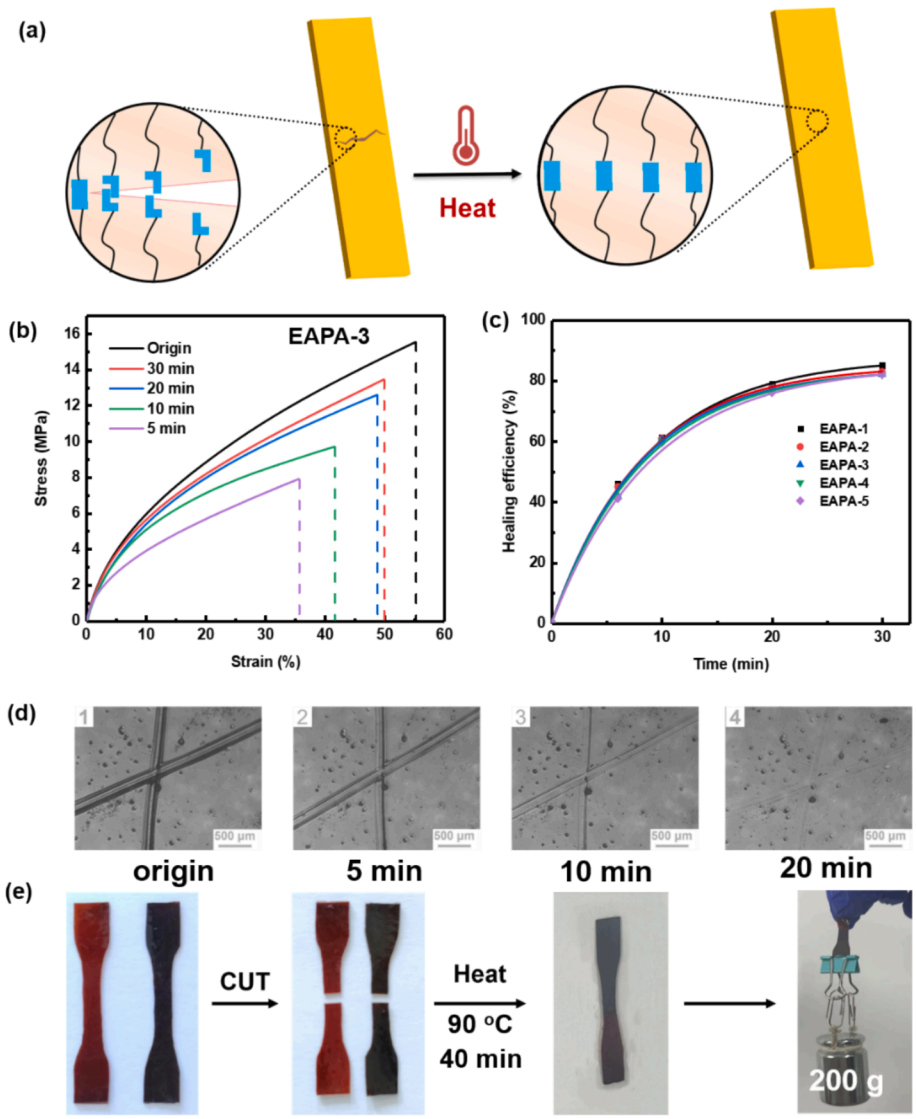


Fig. 4. (a) Schematic of the self-healing process. (b) Stress–strain curves of the healed EAPA-3 film at 70 °C at different times. (c) Healing efficiency as a function of time for EAPA-X films at 70 °C. (d) Time-dependent optical microscopy images of the EAPA-3 film during healing at 70 °C. (e) The experiment showing the EAPA-3 film supporting a weight before and after welding and healing.

Table 3
Self-healing data for EAPA-3 films and other works.

CAN	Healing temperature	Healing efficiency	Healing time
EAPA-3	70 °C	85 %	20 min
SCV-1071d-18[a]	120 °C	–	25 min
ECMP-1[b]	160 °C	70 %	20 min
A6[c]	40 °C	57 %	12 h
Cel-Cp-6[d]	100 °C	80 %	30 min

a [18], *b* [22], *c* [43], *d* [44].

after three recoveries (shown in Fig. S11 and Fig. S12), and the results showed that the thermal properties of EAPA-3 were well preserved.

4. Conclusion

In conclusion, a self-healing and recyclable EAPA-X CAN film with tunable mechanical properties was successfully developed using biobased dialdehydes, biobased amines, and ANN as raw materials. By modulating the ANN content, the mechanical properties of the EAPA-X films were significantly altered. The results showed that the Young's modulus of the EAPA-X films increased by a factor of 63.3 with the incorporation of only 10 % ANN. The activation energy of the EAPA-X films was almost unaffected by the increase in crosslink density, exhibiting very similar activation energies across different films. Despite the high glass transition temperatures, the EAPA-X films demonstrated excellent self-healing properties and minimal mechanical degradation. These findings highlight the superior tunability of the mechanical properties of EAPA-X films, along with their remarkable self-healing and recyclability. Such characteristics make EAPA-X films a promising candidate for sustainable development in the fields of smart materials, electronic packaging, and flexible semiconductor materials.

CRediT authorship contribution statement

Zhiwen Song: Writing – original draft, Formal analysis, Data curation. **Yuequn Ren:** Formal analysis, Data curation. **Shuang He:** Formal analysis. **Yao Liu:** Data curation. **Fei Gao:** Resources, Funding acquisition, Formal analysis, Data curation. **Wang Li:** Formal analysis. **Yujun Cheng:** Formal analysis. **Xuelang Gao:** Formal analysis, Data curation. **Liang Shen:** Funding acquisition, Formal analysis.

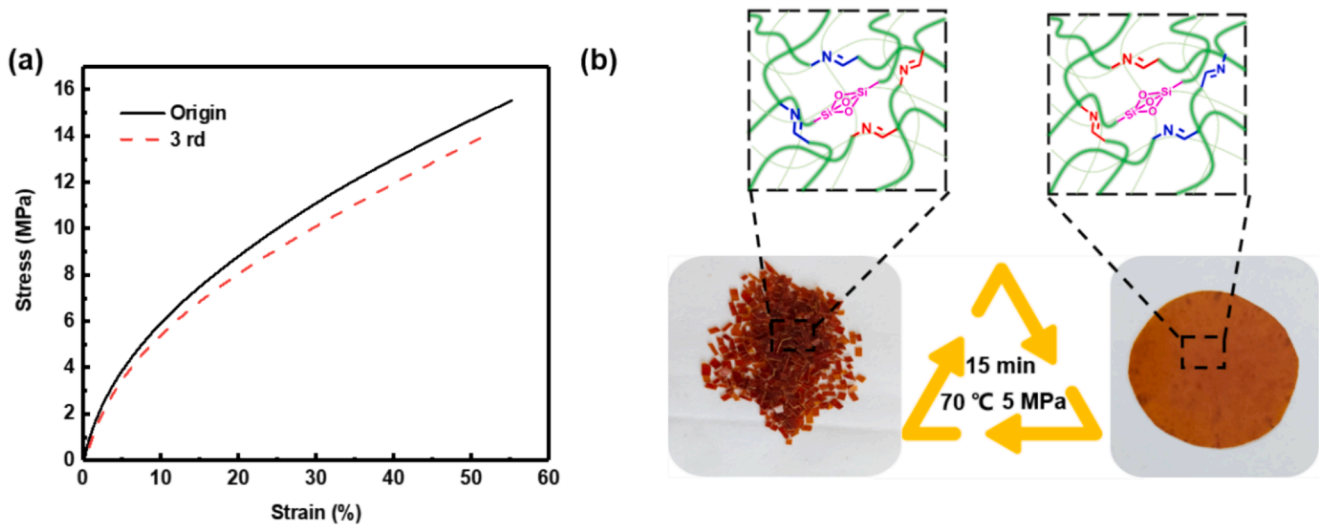


Fig. 5. (a) Tensile curves of the original and remolded EAPA-3 film. (b) Digital images of EAPA-3 film before and after recycling.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113722.

Data availability

No data was used for the research described in the article.

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